

MRTMOS – Week 3

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Disclaimer: This has been written completely using my own knowledge and some sources. AI has only been used to fact-check and write-check at the end.

Agenda

- Which topologies to implement?
 - Which ones to skip?
 - Why?
 - Math + reasoning
- Which elements should be off-chip or on-chip?
 - Why?
 - Math + reasoning

Implementation of Topologies

Assumptions and convention

- For single-input amplifiers, the current gain $G_m = I_{out}/V_{in}$ would apply.
- For 2-input amplifiers, the current gain $G_m = I_{out}/(V_+ - V_-)$ would apply.
 - G_m would adjust and change sign.
 - This is for algebraic simplicity.
 - V_+ is the noninverting terminal of the amplifier, V_- is the inverting terminal of the amplifier.
 - This is under the condition that the amplifier has only one output.
- At every node, KVL and KCL will prevail along with the sign convention.

Schematics for all the stages

- **All Stages and Feedforward: 5-T OTA**
 - We can adjust parameters (width, length, tail current) depending on the use-case!
 - It can be used to invert as well as non-invert!
 - Only 5 transistors needed!
- **Voltage buffer:** Source follower with feedback network
 - Noninverting, reliable buffer

Why we picked them?

- **5-T OTA for all stages and feedforward:**
 - Standardized topology makes layout reuse and verification effortless.
 - Dual inputs allow toggling between inverting and non-inverting.
 - Self-biasing active load and tail current source eliminates complex feedback circuits.
 - Decent gain via high output resistance with low DC headroom.
 - Uniform architecture simplifies much of our work and student's work!
 - Easy to scale output driving current, gain etc. by adjusting transistor parameters.
- **Common drain amplifier with feedback as voltage buffer:**
 - Insulated gate provides infinite input impedance, protecting stage gain.
 - Blocks the feedforward current path through the Miller cap, killing the RHP zero while preserving the feedback path.
 - Adjusting buffer bias current directly alters g_m , letting us push the LHP zero precisely over the non-dominant pole.

Some alternatives that came to our mind...

- **Common Source Amplifier for stage 3:**
 - **Temptation:**
 - **Maximum slew rate:** Lacks a tail current bottleneck, pulling massive transient surges directly into the capacitive load.
 - **Ultimate voltage swing:** Only two stacked transistors between supply rails allows near rail-to-rail output.
 - **Why 5-T OTA wins over it?**
 - **Polarity lock-in:** Common Source is permanently inverting, forcing messy phase-swapping logic at Stage 1 for RNMC loops.
 - **Topological flexibility:** We gladly trade a few millivolts of swing for dual input gates to simplify the circuit.
- **Folded Cascode OTA (Cascode based circuits in general):**
 - **Temptation:**
 - **Massive single-stage gain:** Stacking transistors multiplies output resistance ($A_v \approx g_m r_o^2$), for the middle stage (the engine of the 3-stage amplifier).
 - **Superior input range:** Allows input signals to swing close to the rails without pushing devices out of saturation.
 - **Why 5-T OTA wins over it?**
 - **Headroom starvation:** Stacking 4–5 transistors vertically consumes too much DC voltage for modern low-voltage rails.
 - **Biasing fragility:** Requires intricate external bias networks; which are very delicate and difficult to manage, require extensive feedback networks.

Some alternatives that came to our mind...

- **Resistive load common source differential pair with tail current**
 - **Temptation:**
 - **Absolute simplicity:** Replaces active PMOS mirrors with basic polysilicon resistors, eliminating mirror pole complexities.
 - **Perfect linearity:** Resistors provide frequency-independent, highly linear load impedance.
 - **Why 5-T OTA wins over it?**
 - **Ohm's law trap:** Pushing Stage 3's 1 mA tail current through a 20 $k\Omega$ resistor demands 20 V of headroom. Impossible on our chip. If we reduce the current, it would place a bottleneck on the slew rate. Shrinking the resistor to fit a 3.3 V supply drops output impedance so low that voltage gain vanishes.
- **Current load common source differential pair with resistance in tail.**
 - **Temptation:**
 - **Built-in linearity:** Tail resistor adds local negative feedback (source degeneration), stabilizing transconductance against process shifts.
 - **No bias distribution:** Eliminates routing a precise DC bias voltage (V_{bias}) to the bottom of the stage.

Some alternatives that came to our mind...

- **Current load common source differential pair with resistance in tail.**
 - **Why 5-T OTA wins over it?**
 - **CMRR destruction:** Good noise rejection requires infinite tail impedance; a physical resistor lets supply noise bleed into the signal.
 - **Throttled drive muscle:** Source degeneration punishes effective transconductance, crippling Stage 3's ability to slew a 1 nF load.
- **Class AB push-pull output:**
 - **Temptation:**
 - **Zero quiescent waste:** Burns virtually zero DC current while resting, but opens floodgates during transient load demands.
 - **Thermal efficiency:** Eliminates burning a continuous, chip-heating 1 mA+ DC tail current just to maintain slew rate.
 - **Why 5-T OTA wins over it?**
 - **Shoot-through risk:** Floating bias networks are temperature-sensitive; thermal drift can turn both devices fully on, shorting the supply to ground.
 - **Single I/O:** We can't connect it in both inverting and noninverting configuration. We need both to realize all our experiments.

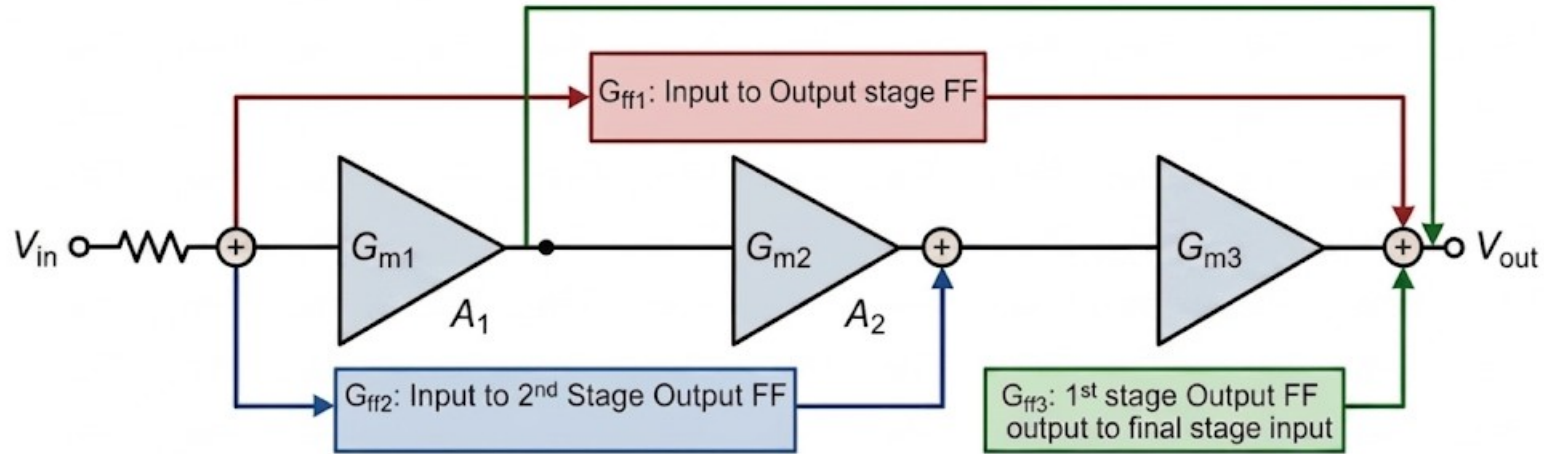
The ones to keep: Standard topologies (No debate!)

- Dominant pole – shunt cap compensation
- Standard Miller compensation
- Miller + nulling resistor
- Nested Miller
- Reverse Nested Miller

The ones that required some analysis

Circuit Diagram

Simplified Multi-Path Feedforward Compensation (Amplifier Stages Only)



Summary (optional text if needed)

Input to Output feedforward

$$\frac{V_{out}}{V_{in}} = \left[\frac{G_{ff} \cdot R_{out,ff}}{1 + \frac{s}{C_{ff} \cdot R_{out,ff}}} + \left(\frac{G_{m1} \cdot R_{out1}}{1 + \frac{s}{C_1 \cdot R_{out1}}} \right) \cdot \left(\frac{G_{m2} \cdot R_{out2}}{1 + \frac{s}{C_2 \cdot R_{out2}}} \right) \cdot \left(\frac{G_{m3} \cdot R_{out3}}{1 + \frac{s}{C_3 \cdot R_{out3}}} \right) \right]$$

- G_{m3} factor would come only if it is 3 stage. Else omit it.
- When you simplify, you'll get s^2 term in numerator. 2 zeroes in 3-stage opamp! 1 zero in 2-stage!
- Calculating the dominant zero (s_{z1}) and non-dominant zeroes (s_{z2}, s_{z3}) (assumption: Main path gain $\gg$ feedforward path gain, 3-stage):

$$s_{z1} \approx -\frac{1}{C_{ff} \cdot R_{out,ff}} \quad s_{z2,3} = \pm j \sqrt{\frac{G_{m1} G_{m2} G_{m3} \cdot C_{ff}}{G_{ff} \cdot C_1 C_2 C_3}} \quad s_{z,2} = \frac{G_{m1} G_{m2} \cdot C_{ff}}{G_{ff} \cdot C_1 C_2}$$

(3-stage) (2-stage)

- **Include it!**

Input node to output of 2nd stage amplifier (Feedforward: 3-stage OTA)

- 3rd stage would add a pole as normal

$$V_2 = \left\{ V_{in} \cdot \left(\left[\frac{G_{m1} \cdot R_{out1}}{1 + s \cdot C_1 \cdot R_{out1}} \right] \cdot G_{m2} \right) + V_{in} \cdot G_{ff} \right\} \cdot \left[\frac{R_{out2}}{1 + s \cdot C_2 \cdot R_{out2}} \right]$$

- Calculating the Zero:
$$s = - \frac{G_{m1} \cdot G_{m2} \cdot R_{out1} + G_{ff}}{G_{ff} \cdot C_1 \cdot R_{out1}}$$

- The numerator has a LHP zero provided that G_{m1} , G_{m2} and G_{ff} are in noninverting (positive) configuration.
- G_{ff} should NOT be negative, else the zero would be propelled to the RHP.
- If it is so, either one of G_{m1} , G_{m2} should be positive but the product of both should be negative, to make the numerator negative as well.

Input of 2nd stage amplifier to output (Feedforward: 3-stage OTA)

- Output node:

$$V_{OUT} = \left\{ V_1 \cdot \left(\left[\frac{G_{m2} \cdot R_{out2}}{1 + s \cdot C_2 \cdot R_{out2}} \right] \cdot G_{m3} \right) + V_1 \cdot G_{ff} \right\} \cdot \left[\frac{R_{out3}}{1 + s \cdot C_3 \cdot R_{out3}} \right]$$

- Calculating the Zero:

$$s = - \frac{G_{m2} \cdot G_{m3} \cdot R_{out2} + G_{ff}}{G_{ff} \cdot C_2 \cdot R_{out2}}$$

- The numerator has a LHP zero provided that G_{m2} , G_{m3} and G_{ff} are in noninverting (positive) configuration.
- G_{ff} should NOT be negative, else the zero would be propelled to the RHP.
- If it is so, either one of G_{m2} , G_{m3} should be positive but the product of both should be negative, to make the numerator negative as well.

Gm followed by Miller in series (2-stage amplifier)

$$V_1 : G_{m1}V_{in} + V_1 \left(\frac{1}{R_{out1}} + sC_1 \right) = 0$$

$$V_x : G_{mff}V_1 = V_x \left(\frac{1}{R_{outff}} + sC_{ff} \right) + (V_x - V_{out})sC_m$$

$$V_{out} : (V_x - V_{out})sC_m = G_{m2}V_1 + V_{out} \left(\frac{1}{R_{out2}} + sC_2 \right)$$

- Solving the KCL equations above, the transfer function and the dominant pole, RHP zero:

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{G_{m1}G_{m2}R_{out1}R_{out2} \cdot \left(1 - sR_{outff} \left[\frac{G_{mff}}{G_{m2}}C_m - C_{ff} - C_m \right] \right)}{(1 + sR_{out1}C_1) \cdot (1 + s(R_{outff}(C_{ff} + C_m) + R_{out2}(C_2 + C_m)) + s^2(R_{out2}R_{outff}(C_2C_{ff} + C_2C_m + C_{ff}C_m)))}$$

$$z_1 = \frac{1}{R_{outff} \left(C_{ff} + C_m \left[1 - \frac{G_{mff}}{G_{m2}} \right] \right)} \quad p_1 = -\frac{1}{R_{out1}C_1}$$

$$p_2 \approx -\frac{1}{R_{outff}(C_{ff} + C_m) + R_{out2}(C_2 + C_m)} \quad p_3 \approx -\frac{R_{outff}(C_{ff} + C_m) + R_{out2}(C_2 + C_m)}{R_{out2}R_{outff}(C_2C_{ff} + C_2C_m + C_{ff}C_m)}$$

- The value of the dominant pole is independent of C_m unlike in normal Miller compensation!
- The value of RHP zero can be altered independent of dominant pole, by altering G_{mff} ! We can push it into LHP!

Capacitor following Gm - Nested Miller (3-stage amplifier)

- Similar thing as 2-stage Miller
- The crux is that Miller introduces an RHP zero and shifts the nondominant pole, if it has a Gm before it. The dominant pole is invariant.
- The magnitude would scale, but the underlying logic is the same.
- There is a potential to shift the zero to LHP!

Should R/C be on/off chip?

Resistor

- Nulling resistors must be placed off-chip.
- Students sweep to empirically discover the exact value.
- On-chip resistors lack any tuning capabilities.
- Wiper resistance uncertainty stays below 5% worst-case.
- Fixed precision metal-film resistors serve secondary verification.
- Precision resistors provide an ultra-tight 0.1% tolerance.
- On-chip resistor variations degrade matching by 20%.
- Off-chip components close the hand-analysis loop.
- Another option: Potentiometers enable continuous sweeping of R_{null} .

Capacitor

- All compensation capacitors must be off-chip components.
- On-chip MOM density is roughly $2\text{fF}/\mu\text{m}^2$.
- Target $C_c = G_{m1}/(2\pi \cdot \text{GBW}) \approx 318 \text{ pF}$.
- Area needed is $159,000\mu\text{m}^2$ per capacitor.
- Total available chip area is $600,000\mu\text{m}^2$.
- On-chip integration is physically completely impossible given that there are other components as well.
- Off-chip C0G dielectric guarantees $\pm 0.1\%$ tolerance.
- On-chip process variations suffer a $\pm 20\%$ error.
- Precise values allow exact experimental verification.

The final verdict

- Keep it off-chip!
- Let students decide the values and plug them in!

A Problem: Breadboard parasitics

- Breadboards are notoriously hostile to high-frequency analog integrated circuits.
- The physical structure of a standard breadboard introduces roughly 2 pF to 5 pF of hidden parasitic capacitance between adjacent rows.
- Furthermore, standard jumper wires act as tiny antennas, introducing stray inductance and coupling ambient noise directly into sensitive paths.
- How does this jeopardize our compensation experiments?
- These invisible environmental parasitics artificially shift the dominant and non-dominant poles before a student even plugs in a compensation capacitor.
- An intentionally uncompensated, unstable OTA might not oscillate simply because the breadboard itself is accidentally acting as a compensation network!
- High-impedance internal nodes routed off-chip will suffer severe phase margin degradation entirely due to the test setup.
- The exact, predictable mathematical derivations we want students to empirically verify (like precisely shifting the RHP zero) become completely muddled by environmental noise.
- **The bottom line: Students must measure the exact physics of our silicon, not the physical flaws of a plastic breadboard.**

Solution: PCB implementation (microcontroller-style board)

- We will bypass the breadboard entirely and design a fully dedicated, professional-grade evaluation PCB.
- This shifts the project from a fragile "student chip" into a robust, self-contained educational microcontroller-style platform.
- To tackle the parasitic capacitance, we must execute a careful ground plane strategy.
- The Trap: Completely removing the ground plane beneath the entire SoC is dangerous; it destroys the return paths for digital SPI control logic, causing digital switching noise to radiate directly into our analog stages.
- The Real Solution: We will maintain a solid ground plane for power delivery and digital logic, but utilize highly localized "copper keep-outs" (voids).
- We will strictly remove all ground and power planes only directly beneath the sensitive high-impedance routing traces and the female plug-in receptacles.
- **We will use our knowledge and experience in PCB design and routing :)**

Hardware and pedagogical features

- For the off-chip component interface, standard Arduino-style square headers are too loose for thin passive leads.
- We will use Machine-Pin (SIP) sockets: Their round, gold-plated inner contacts tightly grip standard COG capacitors and metal-film resistors, guaranteeing near-zero contact resistance.
- The Silkscreen as a Syllabus: We will draw the simplified op-amp block diagrams and routing paths directly on the top-layer silkscreen of the board.
- By explicitly labeling the component sockets (e.g., " C_c Node", " R_{null} Node") right on the board, the PCB itself acts as the student's lab manual.
- Finally, we will add dedicated, on-board SMA or BNC connectors for signal I/O, ensuring perfectly shielded, low-noise interfacing with university oscilloscopes and function generators.

Lab experiments

Sequence and Methodology

- **Phase 1: Single-Stage Characterization (Small & Large Signal)**
 - Measure discrete amplifier blocks
 - Small signal to measure G_m , Large signal to measure swing and ICMR
 - Tail current to measure power ($P_{diss} = V_{DD} * I_{tail}$)
- **Phase 2: Frequency response, compensation (100mV p-p input)**
 - Compare frequency response in uncompensated and various forms of compensation
 - Vary values of R/C or feedforward G_m in compensation networks, and plot graphs to develop intuition.
- **Phase 3: Characterizing multi-stage amplifiers (small & large signal)**
 - Measure the entire amplifier combined, steps similar to phase 1
- **Phase 4: Basic applications and system level applications**
 - Filter design

Measuring and Characterization of single OTA: Gain and output impedance

- Students would connect the amplifier in negative feedback network configuration. Input voltage should be small (50mV peak-peak or so) at a reasonable frequency (5-10kHz).
- The OTA would be connected to a resistive load, and the voltage induced in the resistor would be fed back into the inverting terminal of the OTA. Input voltage would be connected to the noninverting terminal.
- By varying the resistive load R_L , students would in turn, vary the negative feedback gain.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{G_m \cdot R_L || R_{out}}{1 + G_m \cdot R_L || R_{out}} \implies \frac{1}{A_v} - 1 = \left(\frac{1}{G_m} \right) \frac{1}{R_L} + \frac{1}{G_m R_{out}}$$

- Plotting $(1/A_v) - 1$ vs $1/R_L$ yields a straight line where the slope directly equals $1/G_m$. The y-intercept comes out to be $1/G_m \cdot R_{out}$.
- It is recommended to use small values of resistances (upto a few $k\Omega$) to first measure G_m as a small value of R_L would also push the output pole far away.
- At high values of R_L , we can end up measuring R_{out} !
- But at very high values of R_L , the output pole would occur at lower frequencies.
- After experiment is done, students would get the passive and transistor parameters and topologies of amplifiers, and measure via hand calculation!

Measuring Gain and Characterization of the Voltage buffer

- It can be done in a straightforward way, as the voltage gain of the voltage buffer is constant and there is no fear of excess voltage and voltage gain deterioration depending on the load unlike in an OTA.
- Connect a sinusoidal voltage source at the input of the voltage buffer. 50mV peak-peak
- We can measure the output voltage and calculate transfer function.
- We can vary the frequency and amplitude to get an idea of the output swing and bandwidth.
- We can also get to know the ICMR, as we increase the input voltage and we can get to know deviation in output waveform (eg. Clipping) to identify ICMR limits.
- After experiment is done, students would get the passive and transistor parameters and topologies of amplifiers, and measure via hand calculation!

Compensation Networks – Analysis and comparison

- 1) Establish the Baseline:** Students utilize their previously extracted G_m and R_{out} values for the individual blocks.
- 2) The Uncompensated Benchmark:** Students cascade the blocks into a 2- or 3-stage amplifier and measure the raw frequency response (identifying poles, zeros, DC gain, and phase/gain margins).
- 3) Calculate, Then Connect:** Students mathematically design various compensation networks by hand before physically wiring them on the bench. Make the measurements as in step 2.
- 4) Observe and Compare:** Students populate a master observation table to directly compare how each compensation topology shifts the parameters and stabilizes the system.

Analysis and Characterization of multi-stage amplifiers

- Students would connect the multi-stage OTA in negative feedback network configuration. Input voltage should be small (50mV peak-peak or so) at a reasonable frequency (5-10kHz). **Don't connect any compensation network yet for baseline measurements.**
- The OTA would be connected to a resistive load, and the voltage induced in the resistor would be fed back into the inverting terminal of the OTA. Input voltage would be connected to the noninverting terminal.

- By varying the resistive load R_L , students would in turn, vary the negative feedback gain.

$$A_v = \frac{V_{out}}{V_{in}} = \frac{G_m \cdot R_L || R_{out}}{1 + G_m \cdot R_L || R_{out}} \implies \frac{1}{A_v} - 1 = \left(\frac{1}{G_m} \right) \frac{1}{R_L} + \frac{1}{G_m R_{out}}$$

- Plotting $(1/A_v) - 1$ vs $1/R_L$ yields a straight line where the slope directly equals $1/G_m$. The y-intercept comes out to be $1/G_m \cdot R_{out}$.
- It is recommended to use small values of resistances (upto a few $k\Omega$) to first measure G_m as a small value of R_L would also push the output pole far away.
- Compare the experimentally extracted overall parameters (G_m, R_{out}) of the complete multi-stage amplifier against the individual stage parameters previously measured in phase one. Verify the theoretical mathematical relationships of the cascaded stages.

Vary compensation network elements, develop intuition

- **Calculate by Hand:** Derive the transfer function A_v to map out target pole and zero locations.
- **Connect the Topology:** Link the multi-stage amplifier network to your chosen compensation loop configuration.
- **Inject the Step Response:** Vary R,C and feedforward G_m values, and apply a step response to observe the resulting waveforms, and observe the I/O plots and make an observation table.
- **The Transient-to-Control Theory Map:**
 - **Sluggish Rise (No Overshoot):** Overdamped response ($\zeta > 1$) with excessive Phase Margin ($\varphi_m > 76^\circ$) but choked bandwidth.
 - **Ring & Peak Overshoot:** Underdamped condition ($0 < \zeta < 1$); non-dominant poles have crept inside the loop bandwidth, degrading Phase Margin ($\varphi_m < 60^\circ$ typically).
 - **Sustained Oscillations:** Total system instability where Phase Margin hits 0° and loop poles have migrated into the unstable Right-Half Plane (RHP).
 - **The "Slow Creeping Tail":** Diagnostic signature of a Pole-Zero Doublet; imperfect pole-zero cancellation traps an agonizingly slow exponential time constant in your settling path.
 - **Initial Reverse-Direction "Dip":** Classic Non-Minimum Phase behavior caused by a dominant Right-Half Plane (RHP) Zero momentarily outrunning the main amplifier path.
- **First do the calculations and predict the waveforms, then do it practically. Don't be a simulator monkey!**

System level applications 1: Design a filter!

- You have the specifications of all the amplifiers (discrete and combined!)
- Most amplifiers have isolated terminals.
- Design a filter with 1-2 amplifiers:
 - State variable, Multiple feedback, Sallen Key, Biquad, simple amplifier based active filters.
 - Bandwidth is an important aspect in filter design, and poles/zeros and therefore, compensation play a significant role.
- First calculate the transfer function by hand.
- Then form the circuit and connect.
- After that, you can verify the measurements with experimental results.

System level applications 2: Active inductor!

- You have the specifications of all the amplifiers (discrete and combined!)
- Most amplifiers have isolated terminals.
- Design an active inductor with 1-2 amplifiers:
 - Various literature is available
 - Active inductor input impedance is heavily influenced by G_m of the amplifiers, the capacitance of its output node, which is also a pole.
 - Pole compensation indirectly means impedance alteration!
- First calculate the transfer function by hand.
- Then form the circuit and connect.
- After that, you can verify the measurements with experimental results.

System-Level Design: The Compensation Challenge

(Application based Course project)

- **The Objective**
 - Real-world applications break standard amplifiers. You will be assigned strict system-level specifications. Your task is to diagnose exactly why a standard topology fails under these conditions and implement an advanced compensation strategy to save it.
- **The Engineering Workflow**
 - **Step 1: Diagnose the Failure:** Test a baseline amplifier against your assigned specs to find the exact instability mechanism (e.g., ringing, oscillation, or slow settling).
 - **Step 2: Strategize the Fix:** Select an advanced compensation architecture tailored directly to the failure mechanism.
 - **Step 3: Validate the System:** Prove your newly compensated circuit survives across extreme load conditions and process corners.
- **Target Application Scenarios**
 - **Power & Display Drivers:** Stabilizing the amplifier against massive, unpredictable capacitive loads.
 - **Precision ADC Cores:** Eliminating "slow creeping" settling errors to meet strict high-speed clock deadlines.
 - **RF/Wideband Systems:** Bypassing internal bottlenecks to achieve massive bandwidth without destroying phase margin.

Thank You :)

Questions ?